

SSILP Math

Beginner

NUMBER SYSTEMS, SETS AND ABSOLUTE
VALUES



Learning objectives

- Learn about **different number systems**, including real numbers, rational numbers, and irrational numbers.
- Understand and use **sets**, including their definition, notation, and operations such as union and intersection.
- Understand the concept of **absolute value** as a measure of distance and how to evaluate it.

[Interval notation to set builder notation](#) (Organic Chemistry Tutor)

[Absolute value](#) (Organic Chemistry Tutor)

Beginner level videos for fundamentals

Skip if you already know this material

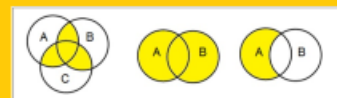
(Note: Some videos may use slightly different notation. Please use the notation in the prescribed textbook for this course)

Session outline

0. Number systems



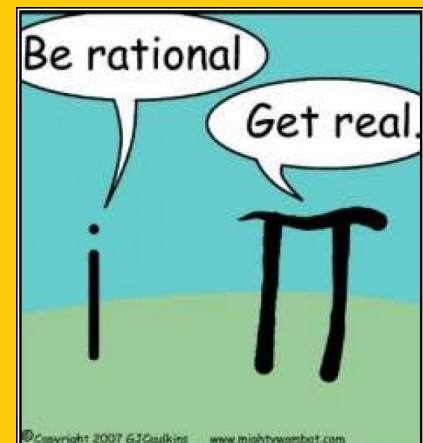
1. Sets and Intervals



2. Absolute value and Distance



0. Number systems



Real numbers

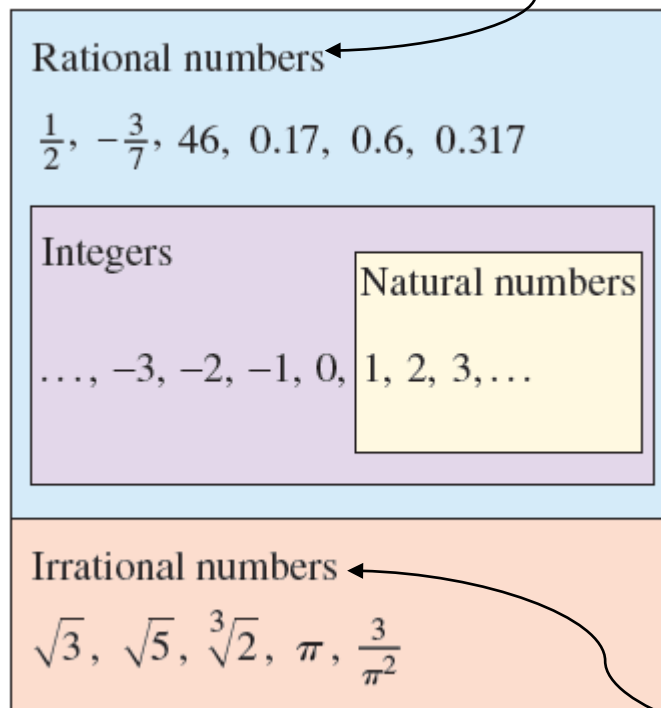
$$\frac{a}{b}$$

$$\frac{1}{3} = 0.\underbrace{3333333333333333}_{\text{repeating pattern}}...$$

This is a **repeating pattern**

the pattern makes this a **rational number**

the set of numbers that
can be expressed as the
ratio of two integers



The real number system

1.4142135623730950488016887...

There is
**no repeating
pattern**

**→
they
continue
forever**

Properties of real numbers

PROPERTIES OF REAL NUMBERS

Property

Example

Description

Commutative Properties

$$a + b = b + a$$

$$7 + 3 = 3 + 7$$

When we add two numbers, order doesn't matter.

$$ab = ba$$

$$3 \cdot 5 = 5 \cdot 3$$

When we multiply two numbers, order doesn't matter.

Associative Properties

$$(a + b) + c = a + (b + c)$$

$$(2 + 4) + 7 = 2 + (4 + 7)$$

When we add three numbers, it doesn't matter which two we add first.

$$(ab)c = a(bc)$$

$$(3 \cdot 7) \cdot 5 = 3 \cdot (7 \cdot 5)$$

When we multiply three numbers, it doesn't matter which two we multiply first.

Distributive Property

$$a(b + c) = ab + ac$$

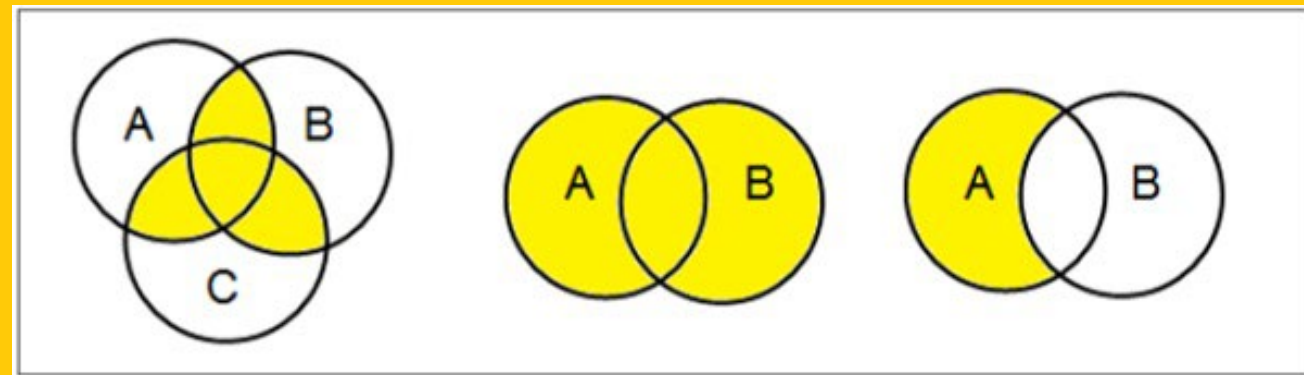
$$2 \cdot (3 + 5) = 2 \cdot 3 + 2 \cdot 5$$

$$(b + c)a = ab + ac$$

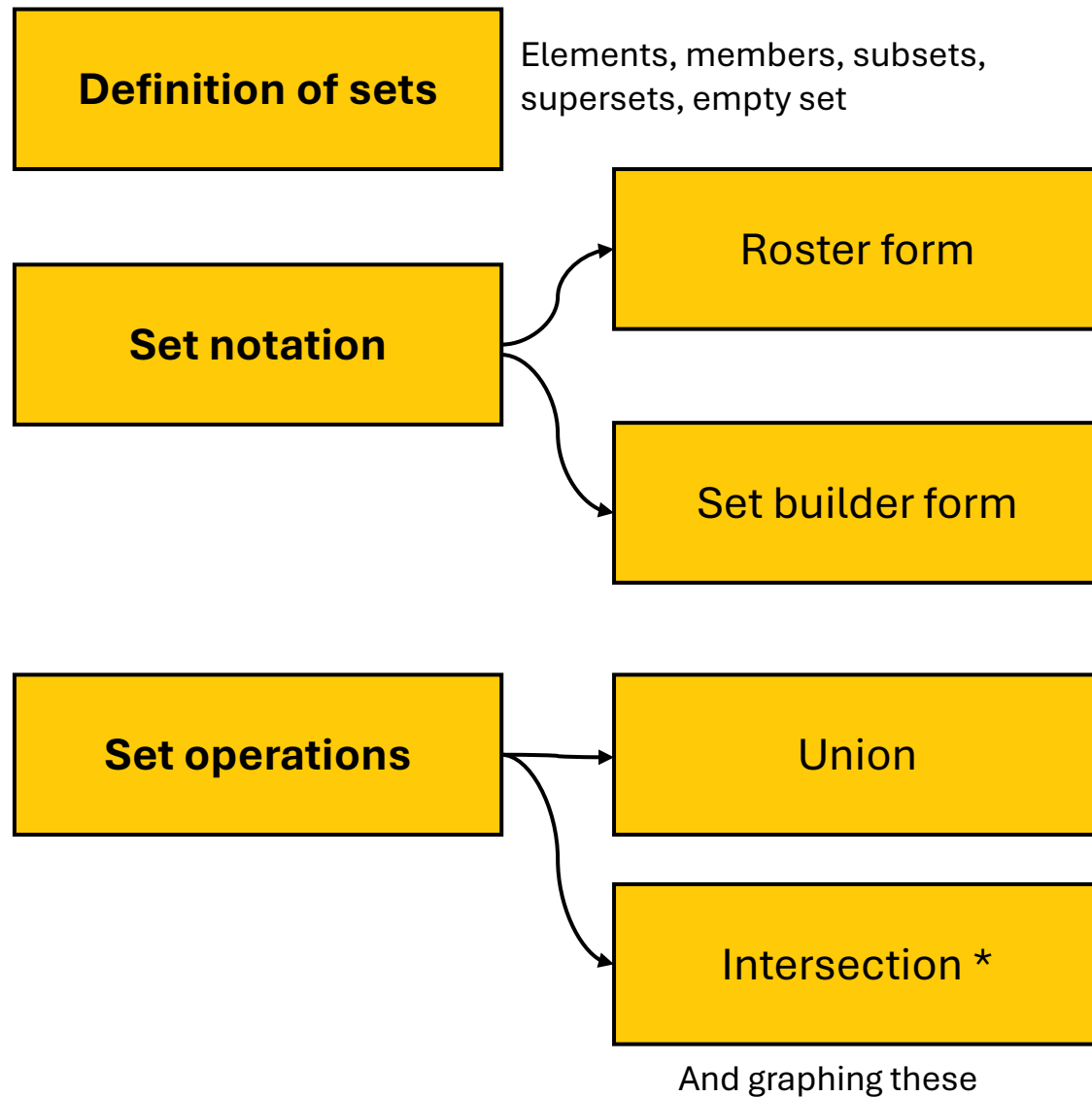
$$(3 + 5) \cdot 2 = 2 \cdot 3 + 2 \cdot 5$$

When we multiply a number by a sum of two numbers, we get the same result as we get if we multiply the number by each of the terms and then add the results.

1. Sets and Intervals



Overview



*Refer to the following
pages in the textbook!*

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 1, Section 1.1:
Real Numbers
Pg 6-8

What is a Set?

A **set** is a collection of objects and these objects are called the elements of the set.

For example, the collection $\{a, b, 1, 2\}$ is a set.

It consists of the collection of four distinct objects, namely, a , b , 1 and 2

If $S = \{a, b, c\}$

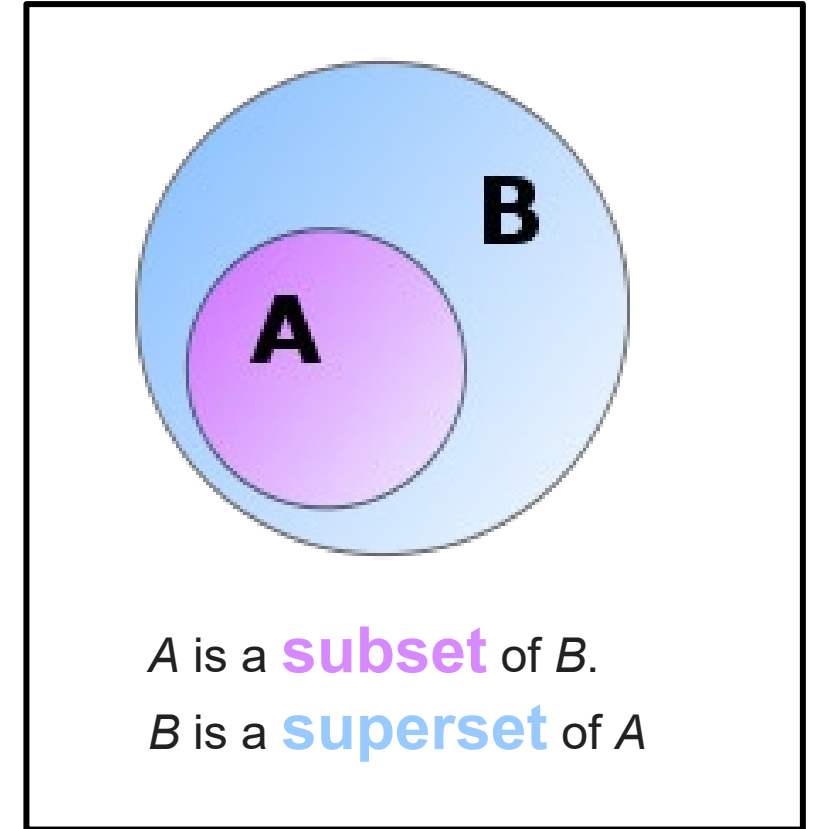
S is a set with elements a, b, c

then $a \in S$

The symbol denotes
“ a is an element of”.

but $d \notin S$

The symbol denotes
“ d is not an element of”



Set Notation

Roster form

Just a few items to list

→ Enclose all items in **curly brackets**

$$\{\text{Miguel, Kristin, Leo, Shanice}\}$$

Long collection of numbers that have a clear pattern

→ use the "..." notation

$$\{3, 6, 9, 12, \dots, 90\}$$

Collection of numbers that have a clear pattern, but **never ends**

→ use the "..." without a number at the end

$$\left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \right\}$$

Set builder form

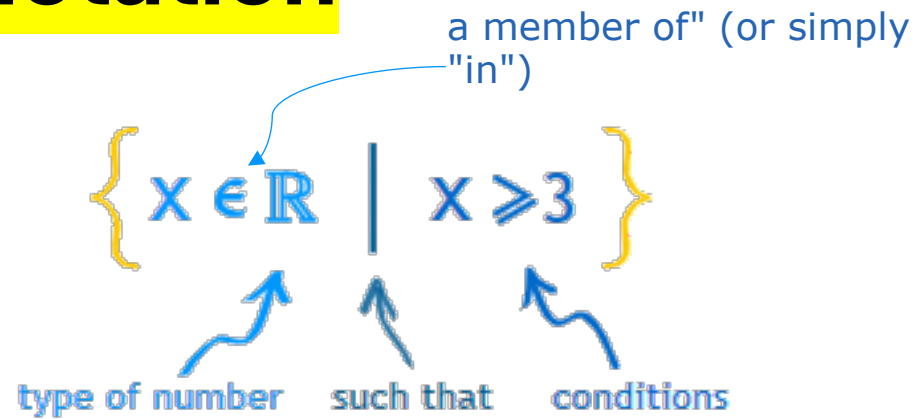
Long collection of numbers that are defined by a rule

→ use the **set builder notation**

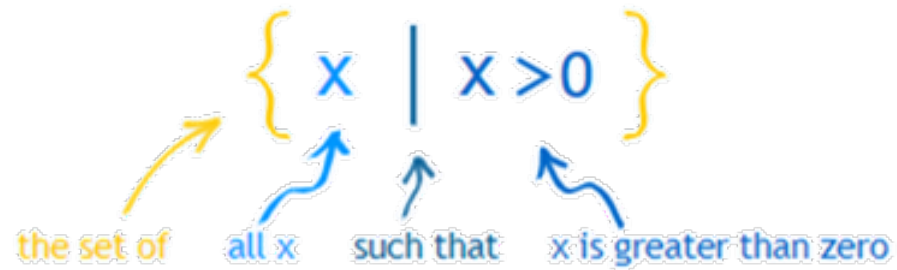
$$\{x \mid x > 18 \text{ and } x \neq 30\}$$

Set builder notation

EXAMPLE 1



EXAMPLE 2



If not mentioned explicitly, it is assumed x is a real number

It says ***"the set of all x's, such that x is greater than 0"***.

In other words **any value greater than 0**

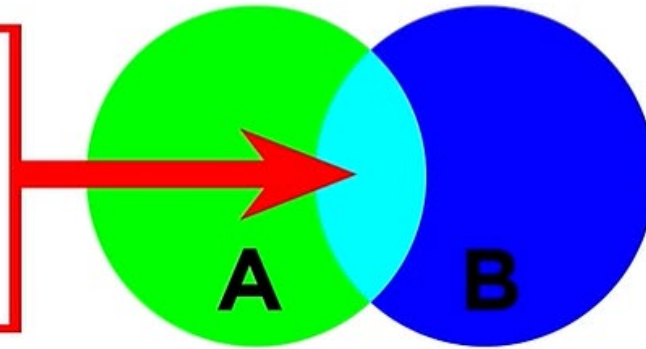
":" can be use instead of |

Set operations: Intersection of sets

$$A = \{1, 4, 5, 9, 13\}$$

$$B = \{2, 5, 6, 8, 9\}$$

the **intersection**
is the set of items
present in both sets



$$A = \{1, 4, 5, 9, 13\}$$

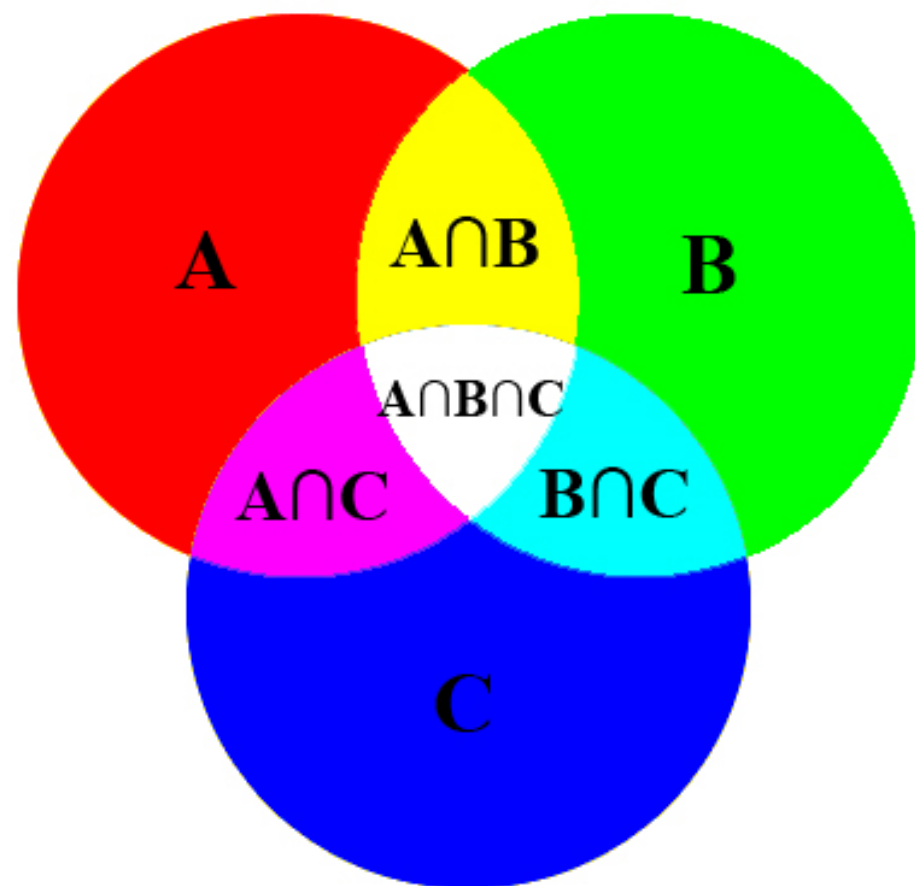
$$B = \{2, 5, 6, 8, 9\}$$

$$A \cap B = \{5, 9\}$$

Intersection symbol ↗

Intersection of three sets

14

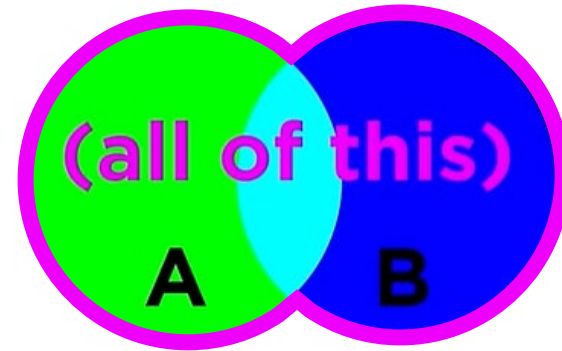


Set operations: Union of sets

$$A = \{1, 4, 5, 9, 13\}$$

$$B = \{2, 5, 6, 8, 9\}$$

the **union** is the set of items present in either of the sets



Elements in Set A
OR
set B
OR
both

$$A = \{1, 4, 5, 9, 13\}$$

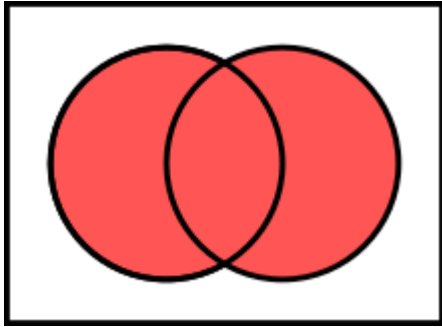
$$B = \{2, 5, 6, 8, 9\}$$

$$A \cup B = \{1, 2, 4, 5, 6, 8, 9, 13\}$$

Union symbol

Union

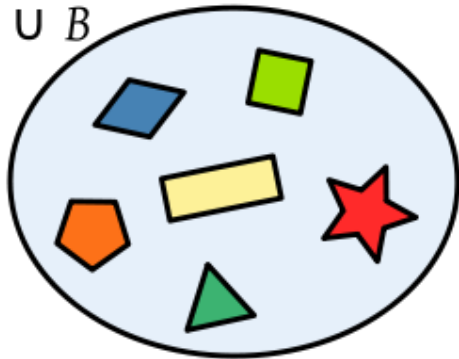
The **union** of A and B , denoted $A \cup B$



$$A = \{\text{orange pentagon}, \text{blue diamond}, \text{green square}, \text{yellow rectangle}\}$$

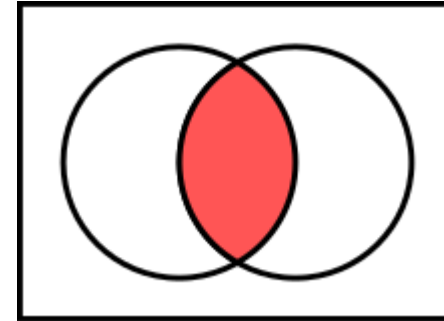
$$B = \{\text{green triangle}, \text{red star}, \text{orange pentagon}\}$$

$$A \cup B$$



Intersection

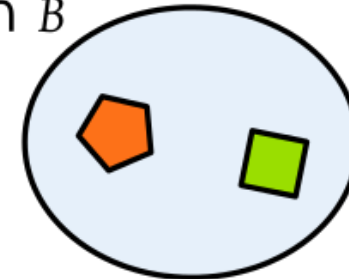
The **intersection** of A and B , denoted $A \cap B$



$$A = \{\text{orange pentagon}, \text{blue diamond}, \text{green square}, \text{yellow rectangle}\}$$

$$B = \{\text{red star}, \text{green square}, \text{green triangle}, \text{orange pentagon}\}$$

$$A \cap B$$



The Null Set

the **null set** is an empty set

$A = \{1, 3, 5, 7, 9, \dots\}$ Set of all odd natural numbers

$B = \{2, 4, 6, 8, 10, \dots\}$ Set of all even natural numbers

$A \cap B = \{ \}$ or $\emptyset$ ← Null set symbol
null set

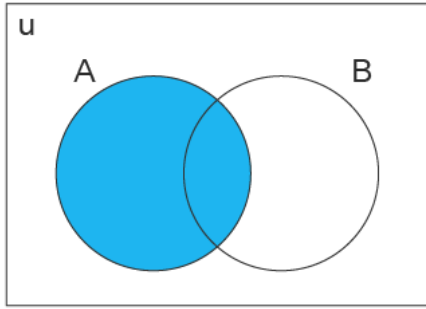


null set $\neq 0$

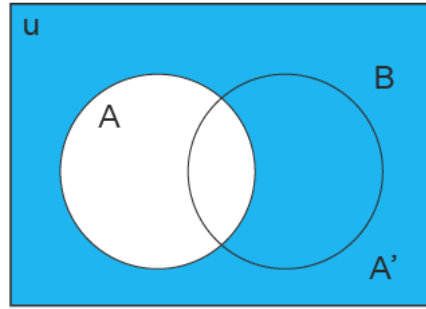
Null Set is **not equal to zero**, because zero is an element that could be included in a set.

The null set means **there is nothing in it**, not even zero.

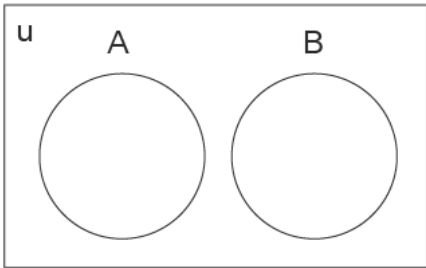
Understanding Set operations using Venn Diagrams



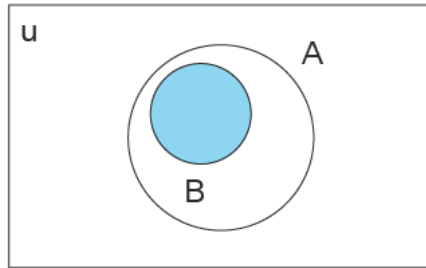
Set A



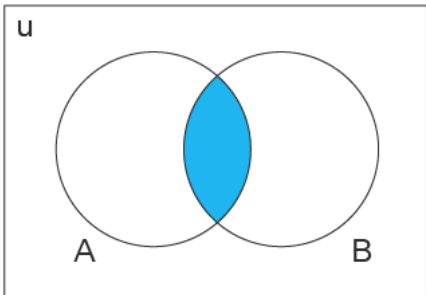
A' is the complement of A



A and B are disjoint sets

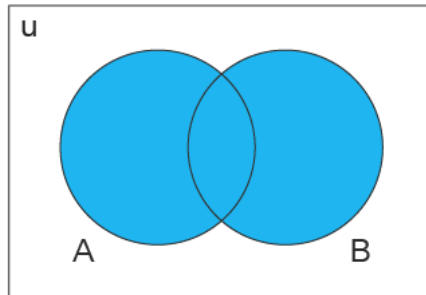


$B \subset A$
B is proper subset of A



$A \cap B$

A and B are overlapping sets



$A \cup B$

A and B are overlapping sets

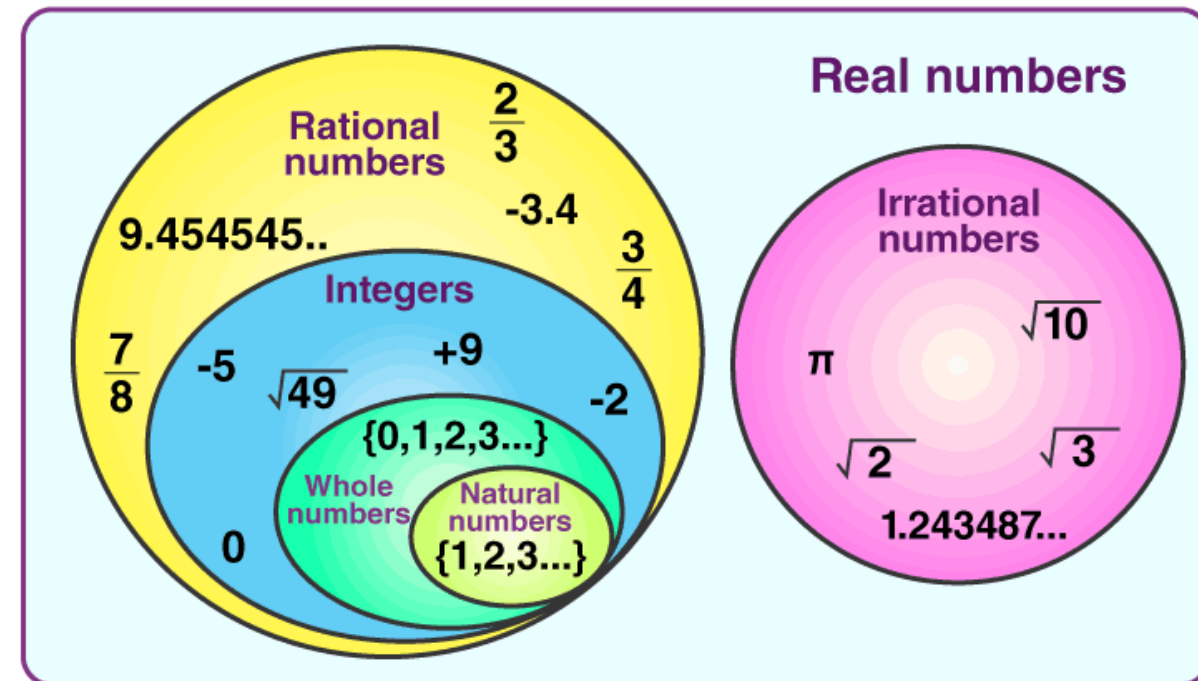
Understanding numbers systems using Sets

Numbers can be classified into sets called **number systems**.

Symbol	Description	Set notation
$\mathbb{N}$	the natural numbers	$\{1, 2, 3, \dots\}$
$\mathbb{Z}$	the integers	$\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
$\mathbb{Q}$	the rational numbers	Ratios of integers: $\left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$
$\mathbb{R}$	the real numbers	an be written using a finite or infinite <i>decimal expansion</i>
$\mathbb{C}$	the complex numbers	These allow us to solve equations such as $x^2 + 1 = 0$

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}.$$

"is a subset of" 



Union and Intersection of Sets

If $S = \{1, 2, 3, 4, 5\}$, $T = \{4, 5, 6, 7\}$, and $V = \{6, 7, 8\}$, find the sets $S \cup T$, $S \cap T$, and $S \cap V$.

SOLUTION

$$S \cup T = \{1, 2, 3, 4, 5, 6, 7\}$$

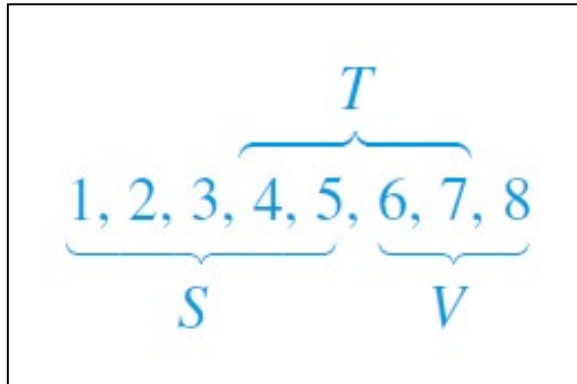
All elements in S or T

$$S \cap T = \{4, 5\}$$

Elements common to both S and T

$$S \cap V = \emptyset$$

S and V have no element in common



EXAMPLE

1.1 Understanding Sets

a) $\{1, 2, 3\} \cup \{3, 4, 5\}$

b) $\{1, 2, 3\} \cap \{3, 4, 5\}$

c) If $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$. Find $A \cup B \cup C$.

d) If $A = \{3, 5, 7, 9, 11\}$, $B = \{7, 9, 11, 13\}$, $C = \{11, 13, 15\}$. Find $A \cap (B \cup C)$.

ACTIVITY 1.1

1.1 Understanding Sets

a) $\{1, 2, 3\} \cup \{3, 4, 5\}$
 $= \{1, 2, 3, 4, 5\}.$

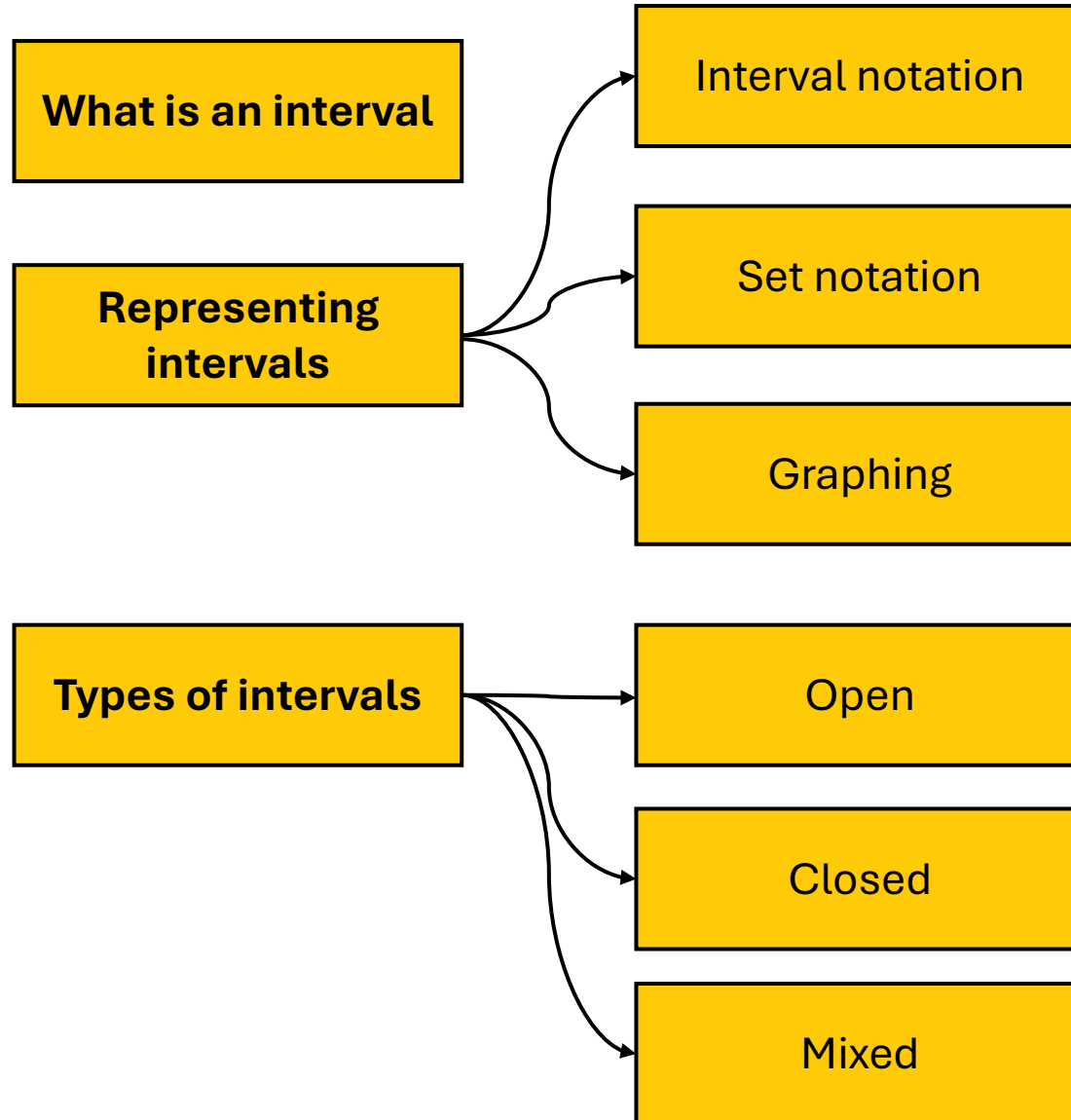
b) $\{1, 2, 3\} \cap \{3, 4, 5\}$
 $= \{3\}.$

c) If $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, $C = \{5, 6, 7, 8\}$.
 $A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 8\}$

d) If $A = \{3, 5, 7, 9, 11\}$, $B = \{7, 9, 11, 13\}$, $C = \{11, 13, 15\}$
 $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 $= \{7, 9, 11\} \cup \{11\}$
 $= \{7, 9, 11\}$

Intervals

Overview



Refer to the textbook for details

Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

Chapter 1, Section 1.1:
Real Numbers
Pg 6-8

What is an interval?

An **interval** is a **range of numbers between two points**.

The two values are called **endpoints** of the interval and *may or may not be included* in the interval.

Open interval



$$(a,b)=\{x\in\mathbb{R} : a < x < b\}.$$

Endpoints Excluded:

consists of all the numbers between a and b , but **does not include** a or b .

Closed interval



$$[a,b]=\{x\in\mathbb{R} \mid a \leq x \leq b\}.$$

Endpoints INCLUDED:

consists of all the numbers between a and b and **including** a and b .

Interval notation

Interval notation is a way to describe continuous sets of real numbers by the numbers that bound them.



Types of intervals

Mixed interval:

Mix of parenthesis and brackets

Infinity is not a number!

It indicates interval has no endpoint but extends infinitely → Use Parenthesis

Notation	Set description	Graph
(a, b)	$\{x \mid a < x < b\}$	
$[a, b]$	$\{x \mid a \leq x \leq b\}$	
$[a, b)$	$\{x \mid a \leq x < b\}$	
$(a, b]$	$\{x \mid a < x \leq b\}$	
(a, ∞)	$\{x \mid a < x\}$	
$[a, \infty)$	$\{x \mid a \leq x\}$	
$(-\infty, b)$	$\{x \mid x < b\}$	
$(-\infty, b]$	$\{x \mid x \leq b\}$	
$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	

Representing Intervals: Examples

$$x > 3$$

solution: $(3, \infty)$



use **parenthesis** because **3 is not part of the solution**

$$x \geq 3$$

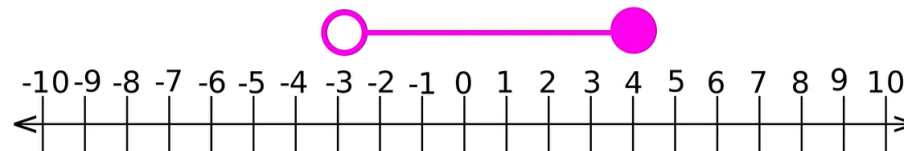
solution: $[3, \infty)$



use **bracket** because **3 is part of the solution**

$$x > -3, x \leq 4$$

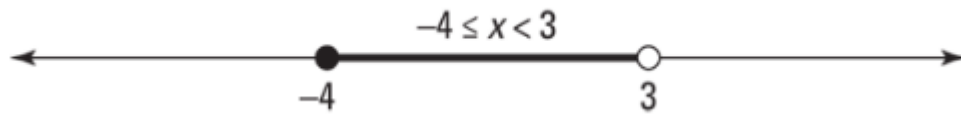
solution: $(-3, 4]$



Graphing Intervals: Example

Graphing Intervals: Express each interval in terms of inequalities, and then graph the interval.

$$[-4, 3)$$



$$\begin{aligned} [-4, 3) &= \{x \mid x \geq -4 \text{ and } x < 3\} \\ &= \{x \mid -4 \leq x < 3\} \end{aligned}$$

This is an example of a **mixed** interval

Unions and Intersections of Intervals

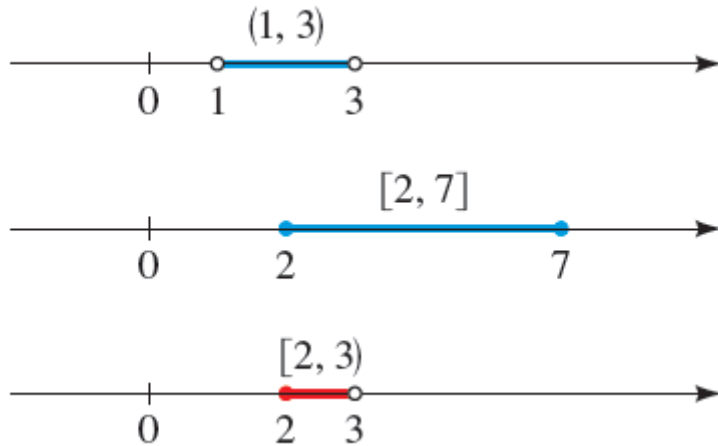
Graph each set.

(a) $(1, 3) \cap [2, 7]$

SOLUTION

(a) The intersection of two intervals consists of the numbers that are in both intervals. Therefore

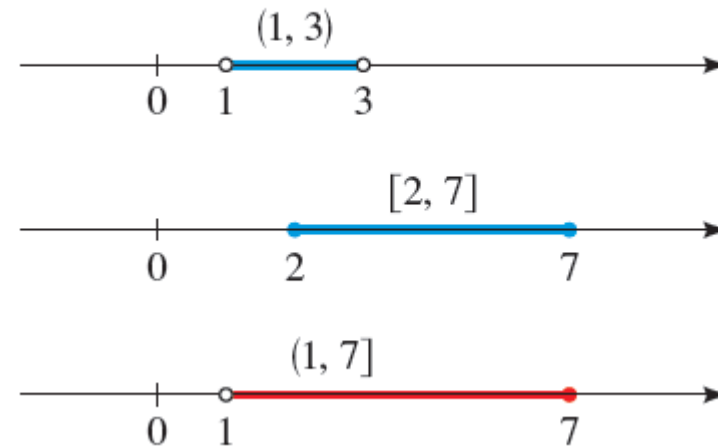
$$\begin{aligned}(1, 3) \cap [2, 7] &= \{x \mid 1 < x < 3 \text{ and } 2 \leq x \leq 7\} \\ &= \{x \mid 2 \leq x < 3\} = [2, 3)\end{aligned}$$



(b) $(1, 3) \cup [2, 7]$

The union of two intervals consists of the numbers that are in either one interval or the other (or both). Therefore

$$\begin{aligned}(1, 3) \cup [2, 7] &= \{x \mid 1 < x < 3 \text{ or } 2 \leq x \leq 7\} \\ &= \{x \mid 1 < x \leq 7\} = (1, 7]\end{aligned}$$



EXAMPLE

1.2. Graphing Intervals

Express each interval in terms of inequalities, and then graph the interval.

a) $(-2, 3)$

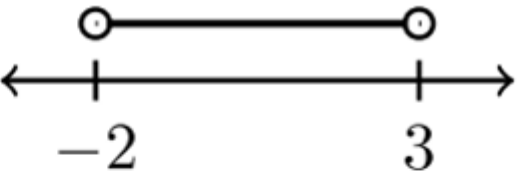
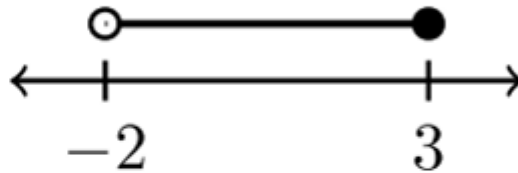
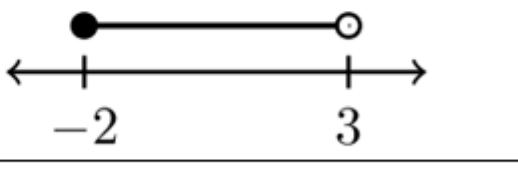
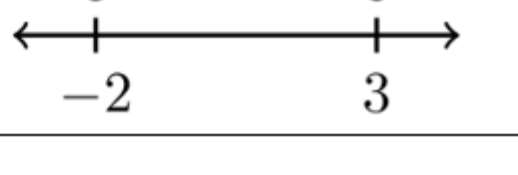
b) $(-2, 3]$

c) $[-2, 3)$

d) $[-2, 3]$

ACTIVITY 1.2

1.2. Graphing Intervals

a	$(-2, 3)$	$\{x \mid -2 < x < 3\}$	
b	$(-2, 3]$	$\{x \mid -2 < x \leq 3\}$	
c	$[-2, 3)$	$\{x \mid -2 \leq x < 3\}$	
d	$[-2, 3]$	$\{x \mid -2 \leq x \leq 3\}$	

1.3 Graphing Intervals: Unions and Intersections of Intervals

Graph the following unions and intersections

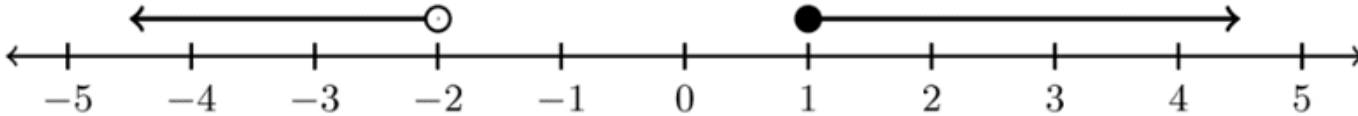
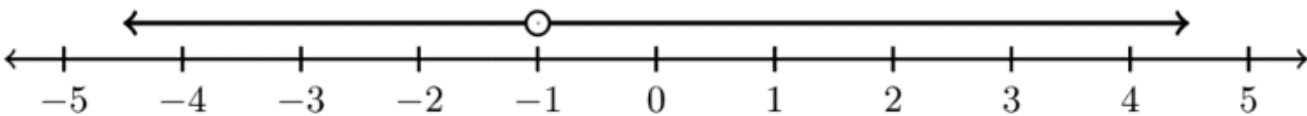
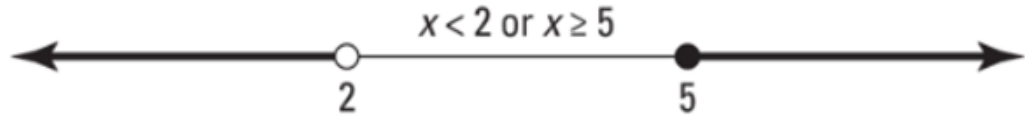
a) $(-\infty, -2) \cup [1, \infty)$

b) $(-\infty, -1) \cup (-1, \infty)$

c) $(-\infty, 2) \cup [5, \infty)$

ACTIVITY 1.3

1.3 Graphing Intervals: Unions and Intersections of Intervals

a)	$(-\infty, -2) \cup [1, \infty)$	 <i>Note this interval has a non-overlapping section!</i>
b)	$(-\infty, -1) \cup (-1, \infty)$	
c)	$(-\infty, 2) \cup [5, \infty)$	 <i>Note this interval has a non-overlapping section!</i>

2. Absolute value and Distance

Are you
positive?

8

Absolutely!

$|-4|$

Absolute value and distance

Overview

Definition

Understanding Absolute value as distance
Understanding Absolute value as a 'piece-wise' function

Properties

Evaluating absolute values of numbers

*Refer to the following
pages in the textbook!*

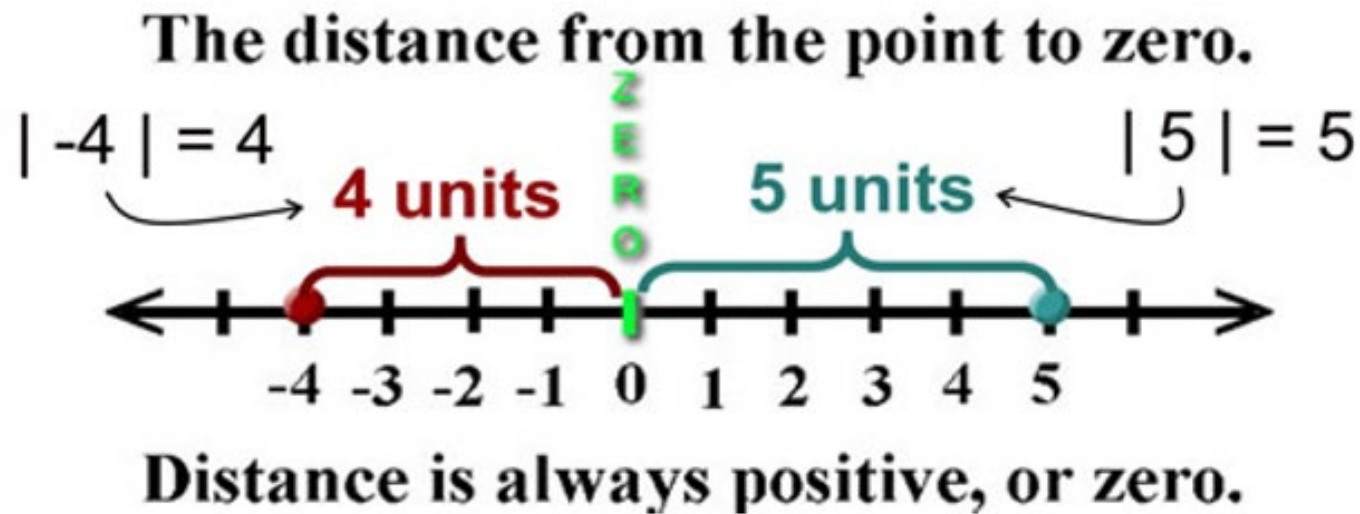
Pre-calculus : Mathematics for Calculus
By Stewart, Redlin, Watson (7th Ed)

**Chapter 1, Section 1.1:
Real Numbers**

Pg 8-9

What is absolute value?

The **absolute value** of a number x is written as $|x|$ and **represents the distance** x is from zero.



Definition of absolute value

DEFINITION OF ABSOLUTE VALUE

If a is a real number, then the **absolute value** of a is

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$



The absolute value does *not* just turn minuses into pluses. That is,

$$|2x - 1| \neq 2x + 1$$

Properties of absolute value

PROPERTIES OF ABSOLUTE VALUE

Property	Example	Description
1. $ a \geq 0$	$ -3 = 3 \geq 0$	The absolute value of a number is always positive or zero.
2. $ a = -a $	$ 5 = -5 $	A number and its negative have the same absolute value.
3. $ ab = a b $	$ -2 \cdot 5 = -2 5 $	The absolute value of a product is the product of the absolute values.
4. $\left \frac{a}{b}\right = \frac{ a }{ b }$	$\left \frac{12}{-3}\right = \frac{ 12 }{ -3 }$	The absolute value of a quotient is the quotient of the absolute values.
5. $ a + b \leq a + b $	$ -3 + 5 \leq -3 + 5 $	Triangle Inequality

Evaluating absolute values of numbers

(a) $|3| = 3$

(b) $|-3| = -(-3) = 3$

(c) $|0| = 0$

(d) $|3 - \pi| = -(3 - \pi) = \pi - 3$ (since $3 < \pi \Rightarrow 3 - \pi < 0$)

EXAMPLE

2.1 Evaluating absolute values of numbers

(a) $|100|$

(b) $|-73|$

(c) $|\sqrt{5} - 5|$

(d) $|10 - \pi|$

ACTIVITY 2.1

2.1 Evaluating absolute values of numbers

Refer to the definition of absolute value!



DEFINITION OF ABSOLUTE VALUE

If a is a real number, then the **absolute value** of a is

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

(a)

$$|100|$$

$$100 > 0$$

$$|100| = 100$$

(b)

$$|-73|$$

$$-73 < 0$$

$$|-73| = -(-73) = 73$$

(c)

$$|\sqrt{5} - 5|$$

$$(\sqrt{5} - 5) < 0$$

$$\begin{aligned} |\sqrt{5} - 5| &= -(\sqrt{5} - 5) \\ &= 5 - \sqrt{5} \end{aligned}$$

(d)

$$|10 - \pi|$$

$$(10 - \pi) > 0$$

$$|10 - \pi| = 10 - \pi$$

2.2 Evaluating absolute values of numbers

$$(a) \quad ||-6| - |-4||$$

$$(b) \quad \frac{-1}{|-1|}$$

$$(c) \quad |2 - |-12||$$

$$(d) \quad -1 - |1 - |-1||$$

$$(e) \quad |(-2) \cdot 6|$$

$$(f) \quad |(-\frac{1}{3})(-15)|$$

$$(g) \quad \left| \frac{-6}{24} \right|$$

$$(h) \quad \left| \frac{7 - 12}{12 - 7} \right|$$

ACTIVITY 2.2

2.2 Evaluating absolute values of numbers

Use PEMDAS!

$$\begin{aligned} \text{(a)} \quad & |{-6} - |{-4}|| \\ &= |6 - 4| \\ &= |2| \\ &= \mathbf{2} \end{aligned}$$

$$\text{(b)} \quad \frac{-1}{|{-1}|} = \frac{-1}{1} = -\mathbf{1}$$

$$\begin{aligned} \text{(c)} \quad & |2 - |{-12}|| \\ &= |2 - 12| \\ &= |{-10}| \\ &= \mathbf{10} \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad & -1 - |1 - |{-1}|| \\ &= -1 - |1 - 1| \\ &= -1 - |0| \\ &= -\mathbf{1} \end{aligned}$$

$$\text{(e)} \quad |(-2)6| = |{-12}| = \mathbf{12}$$

$$\text{(f)} \quad \left| \left(\frac{-1}{3} \right) (-15) \right| = |5| = \mathbf{5}$$

$$\text{(g)} \quad \left| \left(\frac{-6}{24} \right) \right| = \left| \frac{-1}{4} \right| = \frac{\mathbf{1}}{\mathbf{4}}$$

$$\text{(h)} \quad \left| \frac{7 - 12}{12 - 7} \right| = \left| \frac{-5}{5} \right| = |-1| = \mathbf{1}$$

SOLUTIONS